

```

1  n = int(input("Enter number of processes: "))
2  tq = int(input("Enter Time Quantum: "))
3  pid = []
4  at = []
5  bt = []
6  rt = []
7  for i in range(n):
8      pid.append(input(f"Enter PID for P{i+1}: "))
9      at.append(int(input("Arrival Time: ")))
0      bt.append(int(input("Burst Time: ")))
1      rt.append(bt[i])
2
3  ct = [0] * n
4  tat = [0] * n
5  wt = [0] * n
6  time = 0
7  queue = []
8  visited = [0] * n
9  completed = 0
0  while completed < n:
1      # Add processes that have arrived
2      for i in range(n):
3          if at[i] <= time and visited[i] == 0:
4              queue.append(i)
5              visited[i] = 1
6
7      if not queue:
8          time += 1
9          continue

```

```

    time += 1
29     continue
30
31     i = queue.pop(0)
32
33     if rt[i] > tq:
34         time += tq
35         rt[i] -= tq
36     else:
37         time += rt[i]
38         rt[i] = 0
39         ct[i] = time
40         completed += 1
41     for j in range(n):
42         if at[j] <= time and visited[j] == 0:
43             queue.append(j)
44             visited[j] = 1
45     if rt[i] > 0:
46         queue.append(i)
47     for i in range(n):
48         tat[i] = ct[i] - at[i]
49         wt[i] = tat[i] - bt[i]
50     print("\nPID\tAT\tBT\tCT\tTAT\tWT")
51     for i in range(n):
52         print(f"{pid[i]}\t{at[i]}\t{bt[i]}\t{ct[i]}\t{tat[i]}\t{wt[i]}")

```

Enter number of processes: 3

Enter Time Quantum: 5

Enter PID for P1: A

Arrival Time: 1

Burst Time: 9

Enter PID for P2: B

Arrival Time: 2

Burst Time: 16

Enter PID for P3: C

Arrival Time: 3

Burst Time: 12

PID	AT	BT	CT	TAT	WT
A	1	9	20	19	10
B	2	16	38	36	20
C	3	12	37	34	22